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PACKING STEINER FORESTS IN GRAPHS

BY

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THESIS

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Abstract

We study the problem of Steiner Forest packing in a multigraph $G = (V, E)$, where given an input $S_1, S_2, \dots, S_t \subseteq V(G)$, the goal is to find k edge disjoint subgraphs $H_1, H_2, \dots, H_k$, each called a Steiner Forest, such for all $i \in [t], j \in [k]$, S_i is connected in H_j . The specific case of $t = 1$ is the Steiner Tree packing problem, and Kriesell's conjecture postulates that for any $S \subseteq V(G)$, if S is $2k$ -edge-connected then there exist k edge disjoint subgraphs in G such that S is connected in each of them. In his thesis, Lap Chi Lau asserted that if each $S_1, S_2, \dots, S_t$ is $30k$ -edge-connected, then there exist k edge disjoint Steiner Forests. In his proof, he used a theorem called The Extension Theorem. We demonstrate a gap in Lau's proof by providing a counterexample to the Extension Theorem. Furthermore, we resolve this gap by modifying Lau's proof to show that for $k \geq 8$, $35k$ -edge-connectivity suffices to pack k edge-disjoint Steiner Forests.

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Chapter 1

Introduction and Definitions

1.1 Introduction

Given a multigraph $G = (V, E)$, an edge disjoint *packing* $H_1, H_2, \dots, H_k$ is a set of subgraphs $H_i \subseteq G$ where for all $i \neq j \in \{1, 2, \dots, k\}$, $E(H_i) \cap E(H_j) = \emptyset$. Tutte [1] and Nash-Williams [2] gave a characterization of loopless multigraphs which can admit a packing of k edge-disjoint spanning trees, and a quick corollary of this characterization is that all loopless multigraphs that are $2k$ -edge-connected admit a packing of k spanning trees.

A generalization of this is the problem of packing *Steiner Trees*. Kriesell conjectured that given a loopless multigraph G and a set $S \subseteq V(G)$ where S is $2k$ -edge-connected, then there exist k edge disjoint subgraphs $H_1, H_2, \dots, H_k$ where S is connected in each H_i . These subgraphs are referred to as S -subgraphs, and minimal S -subgraphs are called Steiner Trees. Lau [3] proved that if S is $24k$ -edge-connected, a packing of k Steiner Trees exists. West and Wu [4] improved that bound to $6.5k$, and DeVos, McDonald, and Pivotto [5] improved the result to $5k + 4$.

We study a generalization of this problem known as *Steiner Forest Packing*. Given a loopless multigraph G and vertex subsets $S_1, S_2, \dots, S_t \subseteq V(G)$, a $(S_1, S_2, \dots, S_t)$ -subgraph (also referred to as a *Steiner Forest* when edge minimal), is a subgraph H of G such that

for all i , S_i is connected in H . Lau [3] claimed that if for all i , S_i is $30k$ -edge-connected in G , then G contains k edge-disjoint Steiner Forests. However, we found a gap in Lau's proof which this thesis attempts to remedy. We give a proof for a slightly weaker version of Lau's original claim.

Theorem 1.1.1. *Given a loopless multigraph G and vertex subsets $S_1, S_2, \dots, S_t \subseteq V(G)$, if for all $i \in [t]$, S_i is $35k$ -edge-connected in G where $k \geq 8$, then G contains k edge disjoint $(S_1, S_2, \dots, S_t)$ -subgraphs.*

If we drop the assumption that $k \geq 8$, we can prove that each S_i being $36k$ edge-connected is sufficient by using Lemma 3.0.9 in place of Lemma 3.0.7 and following a similar argument throughout the rest of the proof.

1.2 Definitions

We start with the basic definitions in graph theory, which we state here. A *multigraph* $G = (V, E)$ consists of a nonempty and finite set $V(G)$ of *vertices*, and a finite multiset $E(G)$ of unordered pairs of vertices called *edges*. If both vertices in an unordered pair of an edge are the same, the edge is called a *loop* or *self-loop*. When we restrict $E(G)$ to be a set with no repeated elements, then G is called a *graph*. A vertex v is said to be incident to e if $v \in e$, that is, v is one of the endpoints of e . We use $E(v)$ to denote the edges incident to v , and we say that $|E(v)|$ is the degree of v . Vertices u and v are *neighbors* if $uv \in E(G)$. We use $N(v)$ to denote the neighbors of v . A subgraph H of G is a multigraph $H = (V(H), E(H))$, where $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$. Two subgraphs H_1 and H_2 are said to be *edge-disjoint* if $E(H_1) \cap E(H_2) = \emptyset$.

A *path* P from $v_1 \in V(G)$ to $v_k \in V(G)$ in a multigraph G is a collection of vertices $v_1, v_2, \dots, v_k$ such that for every $i \in [k - 1]$, $v_i v_{i+1} \in E(G)$, and $v_i \neq v_j$ for every $i, j \in [k]$ when $i \neq j$. G is said to be connected if for every $u, v \in V(G)$, there exists a path from u to v . For connected multigraphs G , a *spanning tree* T is a minimally connected subgraph of G

where $V(T) = V(G)$. A multigraph is said to be k -edge-connected if G is still connected after the deletion of any $k - 1$ edges. Similarly, we say that two vertices u, v are k -edge-connected if after the deletion of any $k - 1$ edges on G , there still exists a path between u and v . Menger's Theorem states that u, v is k -edge-connected on G if and only if there exist k edge disjoint paths between u and v . A subset $S \subseteq V(G)$ k -edge-connected, if for every distinct $u, v \in S$, u and v are k -edge-connected.

In order to prove our theorem for Steiner Forest packing, we prove a stronger version of the Steiner Tree packing problem which we refer to as the Extension Theorem, which we note is largely based on the original extension theorem that appears in Lau's thesis. To do this, we first define the notion of a *balanced* partition and an *extension*.

Definition 1.2.1. *Given a vertex v , a partition of $E(v) = P_1 \sqcup P_2 \dots \sqcup P_k$ into k parts is balanced if for all i , $|P_i| \geq 2$, where $E(v)$ denotes the edges incident to v .*

Definition 1.2.2. *Given a vertex v , a subpartition of $P_1 \sqcup P_2 \dots \sqcup P_k \subseteq E(v)$ into k parts, where a part is allowed to be the empty set, is called a balanced subpartition if there exists a balanced partition $E(v) = P'_1 \sqcup P'_2 \dots \sqcup P'_k$ such that for all i , $P_i \subseteq P'_i$.*

In other words, a subpartition of v is balanced if it can be extended to a balanced partition of v . If an edge $e \in E(v)$ is part of P_i for some (sub)partition, we say that the (sub)partition assigns e a *label* of i .

Definition 1.2.3. *Given a graph G , a collection of k edge disjoint subgraphs $H_1, H_2, \dots, H_k$ of G , we say that the $H_1, H_2, \dots, H_k$ balance v if the edge subpartition $E(v) \cap H_1, E(v) \cap H_2, \dots, E(v) \cap H_k$ is a balanced edge subpartition of v . We say that $H_1, H_2, \dots, H_k$ balance a subset of vertices $S \subseteq V(G)$ if they balance all vertices in S .*

Definition 1.2.4. *Given a graph G and a vertex v with a subpartition $P_1 \sqcup P_2 \dots \sqcup P_k \subseteq E(v)$ into k parts where each part is allowed to be empty, we say that a collection of k edge-disjoint subgraphs $H_1, H_2, \dots, H_k$ extend v if all of the following criteria are true.*

1. For all i , $P_i \subseteq E(H_i)$.
2. For all i , v is not a cut vertex of H_i .

We will prove a lemma that we will use frequently throughout the proof of the Extension Theorem, which provides motivation for how we define an extension.

Take a graph G , and take some edge cut that splits the graph into components C_1 and C_2 . We form two graphs G_1 and G_2 , where G_1 is formed by taking G and contracting C_2 into a single vertex v_2 . Similarly, we form G_2 by taking G and contracting C_1 into a single vertex v_1 . Note that by construction, $E(v_1) = E(v_2)$, since both edge sets are the (C_1, C_2) -cut of G .

Take some set $S \subseteq V(G)$, and denote $S_1 = S \cap G_1$ and $S_2 = S \cap G_2$. Suppose that we can find k edge disjoint $S_2 \cup v_1$ subgraphs $H_1, H_2, \dots, H_k$ in G_2 . Then, we take an edge subpartition of v_2 in G_1 to be $P_i = H_i \cap E(v_1)$. Suppose there exist k edge disjoint $S_1 \cup v_2$ -subgraphs $H'_1, H'_2, \dots, H'_k$ on G_1 that extend v_2 . Then, we claim the following. For two graphs $H' \subseteq G_1$ and $H \subseteq G_2$, we use $H \uplus H'$ to denote the subgraph induced by $E(H) \cup E(H')$ in G . We refer to this operation as the *edge-union* of two graphs.

Lemma 1.2.1. $H_1 \uplus H'_1, H_2 \uplus H'_2 \dots H_k \uplus H'_k$ are k edge disjoint S -subgraphs of G .

Proof. We first prove that $H_1 \uplus H'_1, H_2 \uplus H'_2 \dots H_k \uplus H'_k$ are edge disjoint. Suppose that there exists some i, j such that $H_i \uplus H'_i$ and $H_j \uplus H'_j$ have a nonempty intersection. Then either $H_i \cap H'_j \neq \emptyset$ or $H_j \cap H'_i \neq \emptyset$, then without loss of generality, assume the former is true. Then, note that since the only edges G_1 and G_2 share are the edges in the (C_1, C_2) cut, this implies that $H_i(v_2) \cap H'_j(v_1)$ is nonempty, where $H(v)$ denotes the edges of H incident to v . However, by definition of extension and our construction, $H_i(v_2) \subseteq H'_i(v_1)$, hence H'_i and H'_j are not edge disjoint and we arrive at a contradiction.

We now claim that $H_i \uplus H'_i$ connects S for any i . Note that $H'_i - v_2$ is connected, by the definition of an extension, and is a subgraph of $H_i \uplus H'_i$, hence for any $a, b \in S_1$, a and b are connected in $H_i \uplus H'_i$. Now suppose $a \in S_1$ and $b \in S_2$. Then, let P_1 be a path from b to v_1 in H_i , let e be the last edge in the path, and note that e is incident to v_1 . Let v' be the

endpoint of e in G_1 that is not v_2 . Then we can find a path P_2 from v' to a on $H'_i - v_2$. Then $P_1 \uplus P_2$ forms a path between a and b on $H_i \uplus H'_i$, hence a and b are connected.

Finally, suppose that $a, b \in S_2$. Take a path P from a to b on H_i . If P does not contain v_1 , then P exists on $H_i \uplus H'_i$ and we are done. Otherwise, if $v_1 \in P$, break P into two parts P_1 and P_3 , where P_1 is the path from a to v_1 along P and $P_3 = P - P_1$. Let e be the last edge of P_1 and e' be the first edge of P_3 . Since both e and e' are incident to v_1 , they appear on G_1 as well, and let $e = uv_2$ and $e' = u'v_2$ on G_1 . Then, there exists a path P_2 from u to u' on $H'_i - v_2$. Note that $P_1 \uplus P_2 \uplus P_3$ forms a path from a to b on $H_i \uplus H'_i$ and a and b are connected. Then this means that S is connected on $H_i \uplus H'_i$, giving us our desired result. $\square$

1.3 Lau's Extension Theorem

In Lau's thesis [3], the following theorem was used to prove the bound of $30k$ for Steiner Forest Packing. Unfortunately, there is a mistake in this theorem which we elaborate on here.

Conjecture 1.3.1 (Lau's Extension Theorem). *Let G be a loopless multigraph, with $S, R \subseteq V(G)$ such that $S \cap R = \emptyset$, S is Qk -edge-connected and for all $r \in R$, the degree of r is at least Qk , where $Q \geq 30$. Given a vertex v and a subpartition $P_k(v)$ of $E(v)$ into k parts, there are k edge-disjoint S -subgraphs that extend v and balance $S \cup R$ if either the following is true.*

1. $v \in S$, $d(v) = Qk$, and $P_k(v)$ is a balanced edge subpartition of $E(v)$.
2. $N(v) \subseteq S \cup R$, $d(v) \leq Qk$, $v \notin S \cup R$, and there is no edge cut of size at most Qk that breaks G into C_1, C_2 , where $S \subseteq C_2$, $v \in C_1$, and $R \cap C_1 \neq \emptyset$.

However, satisfying the second condition is not always sufficient for an extension of k edge-disjoint S -subgraphs to exist. We will disprove the above conjecture with a counterexample. We note that our counterexample is essentially the same as the one given by DeVos, McDonald, and Pivotto [5] in their paper to justify why balancing is needed for the Extension Theorem.

Lemma 1.3.1. *Given any integer $Q \geq 30$, there exists a loopless multigraph G with $S \subseteq V(G)$, Qk -edge-connected, all vertices in $R \subseteq V(G)$ have degree at least Qk and $S \cap R = \emptyset$, a vertex v with a subpartition $P_k(v)$ of $E(v)$ that satisfies the second condition of Lau's Extension Theorem, but does not have k edge disjoint S -subgraphs that extend $P_k(v)$.*

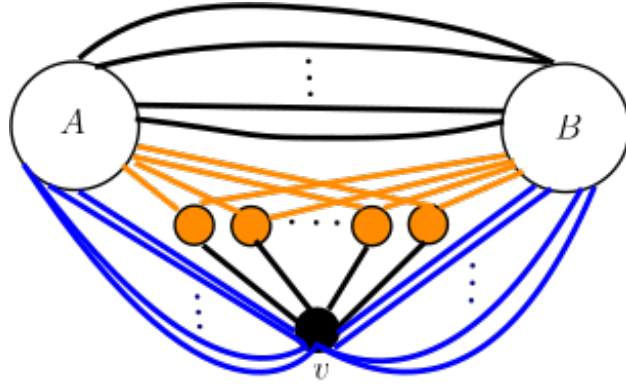
Proof. Take two cliques A, B , both of which are cliques on exactly $Qk + 1$ vertices. Add $Qk - \lfloor (k-1)/2 \rfloor$ edges arbitrarily between A and B , and set $S = A \cup B$. Select $Qk - k + 1$ edges arbitrarily out of the edges between A and B , and denote the set of these edges X , and the set of the other edges between A and B as Y . Subdivide each edge in X , and add a vertex v , which has one edge going to each of the new vertices formed by the subdivision of X . Add $\lfloor (k-1)/2 \rfloor$ edges to v going to vertices in A (not necessarily distinct) arbitrarily, and add $\lceil (k-1)/2 \rceil$ edges to v going to vertices in B (not necessarily distinct) arbitrarily. Add self-loops to the vertices that subdivide X so that all of them have degree at least Qk , and subdivide the self loops themselves to ensure the graph is loopless. Designate all the vertices that subdivide X to be in R . Note that $A \cup B$ is now Qk -edge-connected, as for any $a \in A$ and $b \in B$, $Qk - \lfloor (k-1)/2 \rfloor$ edge disjoint paths can be formed between a and b using $X \cup Y$, and $\lfloor (k-1)/2 \rfloor$ other edge disjoint paths can be formed from a to b by traversing through v . Furthermore, by construction, the degree of v is exactly Qk .

Assume, for contradiction there is a cut of G of size at most Qk into C_1, C_2 such that $v \in C_1$, $S \subseteq C_2$, and $R \cap C_1 \neq \emptyset$. Let $R \cap C_1 = R_1$ and $R \cap C_2 = R_2$. Then, v has $k-1 + |R_2|$ edges incident to a vertex in S or a vertex in R_2 , which all must be part of the cut. Take any vertex in $r \in R_1$, and note that r has two neighbors in S , hence both edges between r and S must also be part of the cut. This means that the cut has at least of $k-1 + |R_2| + 2|R_1| = Qk + |R_1| > Qk$, since R_1 must be nonempty, a contradiction. This means that this graph satisfies the second condition of Lau's Extension Theorem.

For all edges going from v to X , designate them to have a label of 1. For the edges going from v to A , designate them to have labels $2, 3, \dots, \lfloor (k-1)/2 \rfloor + 1$. Finally, for the edges going from v to B , designate them to have labels $\lfloor (k-1)/2 \rfloor + 2, \dots, k$. Note that for any H_i

where $i \neq 1$ to connect A and B , at least one edge in Y must be used, because it is impossible to use the subdivided edges in X , as they must be used in H_1 to connect some $r \in R$ to A or B . However, note that $|Y| = Qk - \lfloor (k-1)/2 \rfloor - (Qk - k + 1) = k - \lfloor (k-1)/2 \rfloor - 1 < k - 1$ as long as $k \geq 3$. Hence, it is impossible to find $H_1, H_2, \dots, H_k$ that extend v , contradicting the theorem of Lau, as desired. $\square$

A figure illustrating the counterexample in the proof above is shown below.



The vertices part of R are highlighted in orange (the subdivided self loops are not pictured). The black edges incident to v have a label of 1, while the blue edges incident to v have labels 2 to k appearing exactly once. The orange edges are the subdivided edges of X in the proof, and the black edges between A and B are the edges in Y . By construction, any subgraph extending a blue edge incident to v must use one of the black edges between A and B to connect the subgraph, since the orange edges need to be used to extend the edges incident to v with a label of 1. However, since by our construction, there are more blue edges than black edges between A and B , such an extension is not possible.

We note that Lau's proof is an induction proof that began by deleting vertices and edges that did not affect the connectivity of S , such as those subdivided self loops in our construction above. Unfortunately, one issue with this approach is that it does not account for the fact that these deletions can make the degree of a vertex in R fall below Qk , hence removing the vertex from R and creating complications when recursing. To remedy this, we give and prove our own version of the Extension Theorem, which we will use to prove a

bound of $35k$ for Steiner Forest Packing.

Theorem 1.3.1 (Extension Theorem). *Suppose there is a multigraph G that may admit self loops, and $S, R \subseteq V(G)$ with $S \cap R = \emptyset$, $|S| \geq 2$, $Q \geq 35$ where Q is an integer, and there exists a vertex $v \in S$ incident to no self loops with $d(v) = Qk$ and a balanced edge subpartition $P_k(v)$. For any $k \geq 8$, if S is Qk -edge-connected in G and every vertex in R is incident to at least $(Q - 2)k$ edges, then there are k edge-disjoint S -subgraphs that balance $S \cup R$ and extend $P_k(v)$.*

If we drop the assumption that $k \geq 8$ we can prove that $Q \geq 36$ is sufficient for the Extension Theorem to hold, by following a similar argument to our proof of the Extension Theorem. Note that while self loops do not affect the connectivity of the graph, it is important that we consider multigraphs which admit self loops, as while recursing down on the graph, we may need to add self loops to some vertices in R to preserve their degree. In lieu of requiring the Extension Theorem to be true for a case similar to the second case in Lau's Extension Theorem, we drop the requirement for a vertex to be in R from being incident to Qk edges to $(Q - 2)k$ edges.

Chapter 2

Steiner Forest Packing

2.1 The Main Theorem

To motivate our Extension Theorem, we demonstrate the proof of Steiner Forest Packing that follows from it. We will proceed by induction, where the following lemma serves as our base case. We will also use methods similar to the proof of this lemma throughout our proof of the Extension Theorem.

Lemma 2.1.1. *Let G be a multigraph that admits self loops. Suppose there is $S, R \subseteq V(G)$ where $S \cap R = \emptyset$, $|S| \geq 2$, S is $35k$ -edge-connected and every vertex in R is incident to at least $33k$ edges. Then if $k \geq 8$, there exist k edge disjoint S -subgraphs that balance $S \cup R$.*

Proof. Delete edges from G until S is minimally $35k$ -edge-connected. If a deleted edge was incident to some $r \in R$, add a self loop to r to preserve the number of edges r is incident to. Then we can find a S -cut of exactly $35k$ edges, and denote the components C_1 and C_2 . Contract C_1 and C_2 into v_1 and v_2 respectively. Let G_1 denote the graph formed by $v_2 + C_1$ and G_2 denote the graph formed by $v_1 + C_2$.

Note that by construction, v_1 has degree $35k$, is incident to no self loops, and is $35k$ -connected to $S \cap C_2$ on G_2 . Hence, we can find k edge disjoint $v_1 \cup (C_2 \cap S)$ -subgraphs that balance $v_1 \cup (C_2 \cap (S \cup R))$ on G_2 by the Extension Theorem. These subgraphs induce a

balanced subpartition at v_2 on G_1 . Note that v_2 is also $35k$ -edge-connected to C_1 , has degree $35k$ and is incident to no self loops. Then we can find k edge disjoint $v_2 \cup (C_1 \cap S)$ -subgraphs that extend v_2 and balance $v_2 \cup (C_1 \cap (S \cup R))$ on G_1 . Then we can take the edge-union of the i th subgraph on G_1 and the i th subgraph on G_2 to give us an S -subgraph on G , hence giving us k edge disjoint S -subgraphs that balance $S \cup R$. Note that we can assume, without loss of generality, that the added self loops were not used in the packing, as self loops do not affect the connectivity and deleting labels at incident edges of a balanced vertex will still keep a vertex balanced. Hence R will still be balanced after we replace the self loops with their original edges. $\square$

Theorem 2.1.1 (Steiner Forest Packing). *Given a loopless multigraph G , $S_1, S_2, \dots, S_t, R \subseteq V(G)$ where $(S_1 \cup S_2, \dots, S_t) \cap R = \emptyset$, such that each S_i is $35k$ -edge-connected, $|S_i| \geq 2$, and all vertices in R are incident to at least $33k$ edges. Then for $k \geq 8$, there exist k edge-disjoint $(S_1, S_2, \dots, S_t)$ -subgraphs that balance R .*

Proof. Suppose, for contradiction, that this is false. Then take a counterexample with $|V(G)| + |E(G)|$ minimized. We can assume that $S_1 \cup S_2 \dots S_t$ is not $35k$ -edge-connected, otherwise this problem reduces down to the previous lemma.

Let C be a subset of $V(G)$ such that the number of edges between C and $V(G) - C$ is at most Qk , and for all $1 \leq i \leq t$, either $S_i \subseteq C$ or $S_i \subseteq V(G) - C$. We know such a C must exist, since there is a $S_1 \cup S_2 \dots S_t$ -cut of less than $35k$, which keeps all S_i intact since it is $35k$ -edge connected. Pick a C with the minimum possible number of vertices, and we will use C_2 to denote $V(G) - C$. Contract C_2 into a vertex v_2 and C into a vertex v to obtain $G_2 = C_2 + v$ and $G_1 = C + v_2$.

If v has degree at least $33k$, add it to R so we can ensure its balancing. Note that since $|C| \geq 2$, G_2 is strictly smaller than G , so we can inductively find k edge disjoint Steiner forests that connect each $S_i \subseteq C_2$ and balance $R \cap C_2$ on G_2 .

Let S^* denote $(S_1 \cup S_2 \dots S_t) \cap C$. We first claim S^* is $35k$ -edge-connected in G_1 . Indeed, if there is a cut of $< 35k$ that splits C into C_a and C_b , it must keep all $S_i \subseteq C$ intact. Without

loss of generality suppose $v_2 \in C_b$, then we can replace C with C_a to get a smaller set that satisfies the desired properties of C , contradicting the minimality of C .

We next claim that there are $d(v_2)$ edge disjoint paths from v_2 to any $s \in S^*$. Suppose, for contradiction, that there was a cut of $< d(v_2)$ with $s \in C_a$ and $v_2 \in C_b$. Then clearly, $S^* \subseteq C_a$, and v_2 cannot be a single vertex since its degree is $d(v_2)$, so $|C_b| \geq 2$. Then this again contradicts the minimality of C . Based on the forest packing on G_2 , copy the corresponding labels over to v_2 on G_1 . Then, pick any vertex $s' \in S^*$, and add $35k - d(v_2)$ fake edges from v_2 to s' .

We claim that v_2 is now $35k$ -edge-connected to any $s \in S^*$. Indeed, suppose there is a cut of $< 35k$ separating v and s . Since the cut is less than $35k$, s' and s will still be connected, so it means that all the fake edges must have been deleted. Then this implies that the deletion of less than $d(v_2)$ edges disconnects v_2 and s , contradicting what we found previously. Assign the fake edges to have no label. Note that if $d(v_2) \geq 33k$, then v_2 is balanced as v is balanced with respect to the packing on G_2 . Otherwise, v_2 is balanced with the addition of the fake edges, since if $d(v_2) < 33k$, we added at least $2k$ fake edges.

Apply the extension theorem to get k -edge-disjoint $S^* \cup v_2$ subgraphs $H_1, H_2, \dots, H_k$ that balances $R \cap C$. We now delete the fake edges we added. Note that since each subgraph H_i is still connected after the deletion of v_2 , all $H_i - v_2$ is still connected after the deletion of the fake edges. Furthermore, since none of the fake edges were incident to a vertex in R , $C \cap R$ is still balanced. Let $H'_1, H'_2, \dots, H'_k$ be the k edge disjoint subgraphs we found on G_2 . We note that $H_1 \uplus H'_1, H_2 \uplus H'_2, \dots, H_k \uplus H'_k$ balances R , are edge disjoint by construction and we claim that $H_i \uplus H'_i$ is a $(S_1, S_2, \dots, S_t)$ -subgraph.

Note that if any $S_j \subseteq C$, then S_j is connected in $H_i \uplus H'_i$ as $H_i - v_2$ is connected. For any $S_j \subseteq C_2$, given $a, b \in S_j$ if there exists a path from a to b that does not traverse through v in H'_i , then a and b are connected on H'_i as well. Otherwise, if all paths from a to b traverse through v in H'_i , take a path P_1 from a to v on H'_i , and a path P_2 from v to b on H'_i . Let e be the last edge of P_1 and e' be the first edge of P_2 . Then by construction, e and e' appear

on H_i (even after deletion of the fake edges). Setting $e = uv_2$ and $e' = u'v_2$, since $H_i - v_2$ is connected, we can find a path from e to e' on $H_i - v_2$ and hence $H_i \uplus H'_i$. Then, we can traverse from a to b on $H_i \uplus H'_i$ by first following the P_1 from a to u , traversing from u to u' , and then following P_2 to reach b . Hence, we have found k edge-disjoint $(S_1, S_2, \dots, S_t)$ -subgraphs that balance R , as desired. $\square$

2.2 When t is small

When $t = 2$ or $t = 3$, we can get a better bound of $9k$ as opposed to $35k$ when Packing Steiner Forests. To prove this, we utilize the Extension Theorem of West and Wu.

Theorem 2.2.1 (West-Wu's Extension Theorem [4]). *Let k and λk be positive integers with $\lambda \geq 6.5$. Suppose there is a loopless multigraph G with some vertex v such that $d(v) = \lambda k$, and $v \in S$ where $S \subseteq V(G)$ is λk -edge-connected. Take an edge subpartition $E_1, E_2, \dots, E_k$ on v into k parts, and let E_0 denote the unlabelled edges incident to v . Then, if $|E_i| \geq 1$ for all $1 \leq i \leq k$ and $|E_0| \geq k$, there exist k edge disjoint S -subgraphs $H_1, H_2, \dots, H_k$ on G that extend v . Moreover, each vertex in S is incident to at least k edges which do not appear in $H_1 \cup H_2, \dots, H_k$, and $(H_1 \cup H_2 \dots H_k) \cap E_0 = \emptyset$.*

Lemma 2.2.1. *Given a graph G , let $S_1, S_2, S_3 \subseteq V(G)$ each be $9k$ -edge-connected in G . Then there exists a packing of k edge-disjoint (S_1, S_2, S_3) -forests.*

Proof. We will first prove this statement when there are two groups, S_1 and S_2 . Note that if $S_1 \cup S_2$ are $7k$ -edge connected, then by West-Wu's Extension Theorem, there exist k edge disjoint Steiner trees that span $S_1 \cup S_2$, giving us our desired result. Otherwise, take the minimum edge cut $G = (G_1, G_2)$ where $S_1 \subseteq G_1$ and $S_2 \subseteq G_2$. Contract G_1 into a vertex v_1 and G_2 into a vertex v_2 . By the extension theorem, we can find k edge-disjoint Steiner trees on $G_2 + v_1$. Take a subpartition $E_1, E_2, \dots, E_k$ of $E(v_2)$ on $G_1 + v_2$, where E_i corresponds to the edges used by the i th tree on $G_2 + v_1$. Note that E_i may be nonempty.

Take any arbitrary $s \in S_1$, and add $9k - d(v_2)$ fake multi-edges from v_2 to s . We claim that $S_1 \cup v_2$ is now $9k$ connected. Indeed, assume there exists an edge cut of size $< 9k$. Note that S_1 will still be connected after such an edge cut, hence if any of the fake edges still exist v_2 will be connected to S_1 . This means that all the fake edges must have been deleted, giving us $< d(v_2)$ edges to disconnect v_2 from S_1 . However, note that if such a cut exists, this would imply the existence of a cut $< d(v_2)$ separating G_1 and G_2 , contradicting the fact that we took the minimum cut. Since $d(v_2) < 7k$, we added at least $2k$ fake edges. For any $1 \leq i \leq k$, if E_i is empty, take a fake edge and add it to E_i , and let $E_0 = E(v_2) - \bigcup_{i=1}^k E_i$, which denotes the unlabeled edges incident to v_2 . Note that $|E_0| \geq k$, and the degree of v_2 is now exactly $9k$. Applying the extension theorem gives us k edge disjoint subgraphs $H_1, H_2, \dots, H_k$ such that $S_1 \cup N_i$, where N_i is the neighborhood given by E_i , is connected for each $H_i - \{v\}$. We can hence delete the fake edges and preserve the connectivity of $S_1 \cup N_i$, and expanding v_2 back into G_2 gives us our desired forest packing.

Now assume there are three groups S_1, S_2, S_3 , each of which are $9k$ -connected. If $S_1 \cup S_2 \cup S_3$ are $7k$ -connected, we are done. Otherwise, take a minimum cut $G = (G_1, G_2)$ that separates $S_1 \cup S_2 \cup S_3$, without loss of generality let $S_1 \subseteq G_1$ and $S_2 \cup S_3 \subseteq G_2$. Contract G_1 into v_1 and G_2 into v_2 . Note that we can pack k Steiner forests on $G_2 + v_1$, and repeating a similar argument as above on $G_1 + v_2$ allows us to extend that packing to S_1 , giving us our desired result.

□

We can extend the ideas from the previous lemma to prove that $(2t+3)k + Ct^2$ connectivity suffices to pack k Steiner trees when there are $t \geq 4$ groups for some universal constant C . This serves as a slightly better bound for the Steiner Forest packing problem when $t < 16$.

Lemma 2.2.2. *There exists an increasing function $f(t) = O(t^2)$, such that for any integer $t \geq 3$ and $k \in \mathbb{N}$, given $S_1, S_2, \dots, S_t$ vertex-disjoint subsets of a graph G where each S_i is $(2t+3)k + f(t)$ -edge-connected, there exists a packing of k edge-disjoint Steiner forests.*

Proof. We will prove this inductively; note that when $t = 3$, $(2t + 3)k = 9k$, and hence a $f(3) = 0$ makes the statement hold. Define $f(t) = f(t - 1) + 2t + 1$ for $t \geq 4$. Let us take $S_1, S_2, \dots, S_t$, and suppose that given $t' < t$ groups in a graph all of which are $(2t' + 3)k + f(t')$ -edge-connected, we can pack k Steiner forests in the graph.

If $S_1 \cup S_2 \dots \cup S_t$ is $7k$ -connected, we are done. Otherwise, take a cut $G = (G_1, G_2)$ of size $\leq (2t+3)k$ such that $|V(G_1)|$ is minimized and at least one terminal vertex exists in G_1 . Contract G_1 and G_2 into v_1 and v_2 respectively. We start by packing $k' = k + \lceil \frac{2}{2t+1}k \rceil$ edge disjoint Steiner forests on $G_2 + v_1$. Let t' be the number of groups in G_2 , and $k' \leq (1 + \frac{2}{2t+1})k + 1$. To prove that we can pack k' edge disjoint Steiner forests, we need to show that $(2t' + 3)k' + f(t') \leq (2t + 3)k + f(t)$, since each group is at least $(2t + 3)k + f(t)$ edge connected on $G_2 + v_1$. We know that $t' \leq t - 1$, hence $(2t' + 3)k' + f(t') \leq (2(t - 1) + 3)((1 + \frac{2}{2t+1})k + 1) + f(t - 1) = (2t + 3)k + f(t)$, meaning that such a packing is possible.

We now prove that all terminal vertices in $G_1 + v_2$ are $(2t + 3)k$ -edge-connected. Indeed, if there is an edge cut of size smaller than $(2t + 3)k$ separating $G_1 + v_2 = (H_1, H_2)$, where H_1 and H_2 both contain terminal vertices, without loss of generality we assume $v_2 \in H_2$. Then this edge cut separates H_1 from $G - H_1$ as well, contradicting the fact that G_1 was minimized. Hence, we can treat all terminal vertices as a single group, which we will call S_1 . Take a vertex $s \in S_1$, and add fake multiedges until the degree of v_2 is exactly $(2t + 3)k$. We claim that $S_1 \cup v_2$ is $(2t + 3)k$ -edge-connected. Suppose we were able to find such a cut (H_1, H_2) of size smaller than $(2t + 3)k$ where $S_1 \subseteq H_1$ and $v_2 \in H_2$. Note that H_2 cannot consist of only v_2 , as there were exactly $(2t + 3)k$ edges (including fake ones) incident to v_2 , and all of them must have been deleted for this to happen. Hence, this means H_1 is strictly smaller than G_1 , giving us an edge cut of at most $(2t + 3)k$ smaller than G_1 , a contradiction.

Let $E_0, E_1, \dots, E_{k + \lceil 2k/(2t+1) \rceil}$ partition $E(v_2)$, where for $1 \leq i \leq k + \lceil 2k/(2t+1) \rceil$, E_i corresponds to the edges used for i th forest on $G_2 + v_1$, and $E_0 = E(v_2) - \bigcup_{i=1}^{k + \lceil 2k/(2t+1) \rceil} E_i$. We will generate $E'_0, E'_1, \dots, E'_k$ to be extended with the following procedure. Re-index $E_1, \dots, E_{k + \lceil 2k/(2t+1) \rceil}$ so that $|E_i| \leq |E_j|$ if $i \leq j$. For $1 \leq i \leq k$, set $E'_i = E_i$, and set

$E'_0 = E_0 + \bigcup_{i=k+1}^{k+\lceil 2k/(2t+1) \rceil} E_i$. Note that $k + \frac{2k}{2t+1} = \frac{2tk+k}{2t+1} + \frac{2k}{2t+1} = \frac{(2t+3)k}{2t+1}$, so a proportion of at least $\frac{2}{2t+3}$ of the edges at v_2 are part of E'_0 . We then note that $|E'_0| \geq (\frac{2}{2t+3})(3+2t)k = 2k$. For any empty E'_i , take an edge from E'_0 and move it to E'_i . Now, note that the degree of v_2 is exactly $(3+2t)k$, $|E'_0| \geq k$, and $E'_1, \dots, E'_k$ are nonempty, hence we can apply the West-Wu Extension Theorem. Deleting the fake edges gives us k Steiner forests, and we are done. $\square$

Chapter 3

The Extension Theorem

We prove the Extension Theorem in this chapter. We note that our proof is roughly similar to Lau's proof of his Extension Theorem.

Theorem 3.0.1 (Extension Theorem). *Suppose there is a multigraph G that may admit self loops, and $S, R \subseteq V(G)$ with $S \cap R = \emptyset$, $|S| \geq 2$, $Q \geq 35$ where Q is an integer, and there exists a vertex $v \in S$ incident to no self loops with $d(v) = Qk$ and a balanced edge subpartition $P_k(v)$. For any $k \geq 8$, if S is Qk -edge-connected in G and every vertex in R is incident to at least $(Q - 2)k$ edges, then there are k edge-disjoint S -subgraphs that balance $S \cup R$ and extend $P_k(v)$.*

Suppose, for contradiction, that this theorem is false. Take a counterexample G where the sum of the vertices and the edges that are not self loops is minimized. We may assume without loss of generality, that each label $1, 2, \dots, k$ appears at least twice on $E(v)$, because we can assign labels to the otherwise unlabelled edges if need be.

Lemma 3.0.1. *No vertex in $V(G) - S - R$ is a cut vertex.*

Proof. Suppose, for contradiction, that there is such a cut vertex, and denote that vertex w . Suppose that the deletion of w splits G into components $C_1, C_2, \dots, C_s$. Add a copy of w to each component, denoted $w_1, w_2, \dots, w_s$, where the edges between w_i and C_i are the edges

between w and C_i in G . If exactly one of these components contains vertices in S , then delete all vertices in the graph not in that component or w , inductively apply the extension theorem - which we may do since at least one vertex was deleted. Note that this automatically gives us k edge disjoint S subgraphs on G that balance $S \cup R$ and extend v . Note the vertices of R not in the same component as S are balanced since none of their incident edges were used in the packing.

If more than one of these components contains vertices in S , then we claim that if C_i contains vertices in s , then $w_i \cup (S \cap C_i)$ is Qk -edge-connected. Indeed, take any vertex in $S \cap C_i$, and take another vertex in S but in some other component C_j . There are at least Qk edge disjoint paths between the two vertices, all of which must pass through w , hence there are Qk edge disjoint paths from any vertex in $S \cap C_i$ to w_i as well. For the component C_i such that $v \in C_i$, we can inductively apply the extension theorem and find k -edge disjoint $S \cup w_i$ -subgraphs that extend v and balance $(C_i \cup w_i) \cap (S \cup R)$, as $C_i \cup w_i$ has strictly fewer vertices and edges than G .

For every other component C_j that contains vertices in S , we wish to find k edge disjoint $S \cup w_j$ subgraphs that balance $(C_j \cup w_j) \cap (S \cup R)$, but existence of subgraphs is not immediately obvious, as we are not guaranteed that there exists some vertex in $(S \cap C_j)$ with degree exactly Qk and hence need to modify the graph to apply the Extension Theorem recursively. We employ a similar strategy as seen in Lemma 2.1.1. To do this, we delete edges from the component until $(S \cap C_j) \cup w_j$ is minimally Qk -edge-connected. For each edge we deleted, if any of its endpoints was in R , add a self loop to that vertex to preserve the number of incident edges. Note that there exists a set of Qk edges that cuts $(S \cap C_j) \cup w_j$, and denote the components of the cut $C_{j,1}$ and $C_{j,2}$. Contract both into one vertex to obtain $C_{j,1} + v_{j,2}$ and $C_{j,2} + v_{j,1}$. Note that $v_{j,1}$ must be Qk -edge-connected to $C_{j,2} \cap (S \cup w_j)$ and $v_{j,2}$ must be Qk -edge-connected to $C_{j,1} \cap (S \cup w_j)$. We can then find k edge disjoint $(C_{j,2} \cap (S \cup w_j)) \cup v_{j,1}$ -subgraphs on $C_{j,2} + v_{j,1}$ that balance $(C_{j,2} \cap (S \cup R \cup w_j)) \cup v_{j,1}$. This induces an edge subpartition on $v_{j,2}$, and we apply the extension theorem on $C_{j,1} + v_{j,2}$ to find k edge disjoint

$(C_{j,1} \cap (S \cup w_j)) \cup v_{j,2}$ -subgraphs on $C_{j,1} + v_{j,2}$ that balance $(C_{j,1} \cap (S \cup R \cup w_j)) \cup v_{j,2}$ and extend $v_{j,2}$. We may assume no self loops were used in these subgraphs, as deleting a self-loop does not affect the connectivity of the graph, and removing the label from the self loop cannot change its respective vertex from a balanced vertex to an unbalanced vertex. Then, taking the edge-union of the i th subgraph on $C_{j,1} + v_{j,2}$ with the i th subgraph on $C_{j,2} + v_{j,1}$ gives us in total k edge disjoint $S \cup w_j$ subgraphs that balance $(C_j \cup w_j) \cap (S \cup R)$. Taking the union of all these subgraphs over each component in G gives us k edge disjoint $S \cup w$ -subgraphs that balance $S \cup w \cup R$, as the existence of w in all $C_1, C_2, \dots, C_s$ ensures connectivity of all of S in each subgraph. $\square$

Theorem 3.0.2 (Mader's Splitting Off Lemma [6]). *Let x be a vertex of a loopless multigraph G . Suppose that x is not a cut vertex and that x is incident with at least 4 edges and adjacent to at least 2 vertices. Then there exists edges xy and xz where $y \neq z$, such that the number of edge disjoint paths between $a, b \in V(G) - x$ in $G' - xy - yz + yz$ is at least the number of edge disjoint paths between a and b in G .*

Lemma 3.0.2. *All $u \in V(G) - S - R$ have degree 3 and are adjacent to 3 vertices.*

Proof. Take a vertex $u \in V(G) - S - R$. We can assume, without loss of generality, that u is not incident to any self loops, because deleting the self loops do not affect the connectivity of S , nor the number of edges each vertex in R is incident to.

If u is incident to only one vertex w , delete it, and note that the deletion does not affect the connectivity of S in $G - u$. If w is a vertex in R , for each edge between u and w , we add a self loop to w corresponding to the deleted edge so that w is still incident to at least Qk edges. We can then inductively find k edge disjoint S -subgraphs that extend v balance $S \cup R$ on $G - u$. If w was in R , we replace each of the added self loops with the corresponding original edge on G , which does not affect the connectivity of any of those S -subgraphs nor the balancing. Note that it is not possible for $w = v$, because if that were true, deleting any edge incident to v would make S no longer Qk -connected, but deleting u and its incident edges would not affect the connectivity of S .

If u is incident with exactly two vertices, y and z , then delete one edge between y and u , and one edge between u and z , and another edge between y and z . Note this does not affect the connectivity of S , nor the degree of any vertex in R , hence we can inductively find k disjoint S -subgraphs that extend v and balance $S \cup R$ on $G - yu - uz + yz$. If the added edge yz appears in one of the S -subgraphs, then delete it and replace it yu and uz , which preserves both connectivity and balancing. Similarly, if u is incident with at least 4 edges, since u is incident to no self loops, then we can delete the self loops from this graph and apply Mader's Splitting Off Lemma at u . We then add the deleted self loops back and follow a similar argument to obtain the desired result.

If u is incident with exactly 3 edges but only has two neighbors, y and z , then note that any path that goes through u has to pass through y and z and use only two edges. Without loss of generality, say that u and y share two edges. Hence, deleting one edge between u and y will not affect the connectivity of S . We can hence delete one edge between u and y , and if $y \in R$ add one self loop to R so it is still incident to at least Qk edges. Note that y cannot be the same as v , because it would contradict one of v having degree Qk and $v \in S$, or $N(v) \subseteq S \cup R$. Hence we can find k edge disjoint S -subgraphs that extend v and balance $S \cup R$ on $G - uy + yy$, and we replace the added self loop with the corresponding edge between u and y on G , giving us our desired result. $\square$

Lemma 3.0.3. *$V(G) - S - R$ is an independent set.*

Proof. Suppose not, and there is an edge e between the vertices in $V(G) - S - R$. If deleting that edge keeps S Qk -edge-connected, then note that such a deletion does not affect any vertices at R nor v , hence we can recurse down to find the S -subgraphs with the desired properties. If S is no longer Qk -edge connected after the deletion of e , then there must be a collection of exactly Qk edges (including e) which cuts S . Suppose that deleting those Qk edges splits G into C_1 and C_2 . Contract C_2 into a single vertex v_2 and C_1 into v_1 . We can assume without loss of generality that $v \in C_1$. Note that the degrees of all vertices are preserved on $C_1 + v_2$ and $C_2 + v_1$. Furthermore, $S \cap C_1$ is Qk -edge-connected in $C_1 + v_2$ and

$S \cap C_2$ is Qk -edge-connected in $C_2 + v_1$.

Since C_2 had to contain one of the endpoints of e and contains a vertex in S , we know that $C_1 + v_2$ is strictly smaller than G . We also note that v_2 must be Qk -connected to $C_1 \cap S$, as there are Qk edge disjoint paths from any vertex in $C_1 \cap S$ to any vertex in $C_2 \cap S$, which all must pass through the edge cut of Qk edges. Hence, we can apply the extension theorem to find k edge disjoint $S \cup v_2$ -subgraphs $S_1, S_2, \dots, S_k$ on $C_1 + v_2$ which extend v and balance $C_1 \cap (S \cup v_2 \cup R)$. This induces a balanced edge subpartition on v_1 , so we apply the extension theorem on $C_2 + v_1$ in a similar manner to get k edge disjoint $S \cup v_1$ -subgraphs $S'_1, S'_2, \dots, S'_k$ which extend v_1 and balance $C_2 \cap (S \cup v_1 \cup R)$ on $C_2 + v_1$. Then, taking $S_1 \cup S'_1, S_2 \cup S'_2, \dots, S_k \cup S'_k$ gives us k edge disjoint S subgraphs on G which extend v and balance $S \cup R$. $\square$

The following lemma allows us to assume, without loss of generality, that every vertex in R has less than $2k$ self-loops.

Lemma 3.0.4. *Let R' be the set of all vertices in R that are incident to less than $2k$ self loops. If there exist k edge disjoint S -subgraphs that extend $P_k(v)$ and balance $S \cup R'$, then there exist k edge disjoint S -subgraphs that extend $P_k(v)$ and balance $S \cup R$.*

Proof. Take a vertex $r \in R - R'$. Suppose there are k edge disjoint S -subgraphs that extend $P_k(v)$ and balance $S \cup R'$. If any self loops incident to r were used in any of the S -subgraphs, we can remove them from the S -subgraph, and note that their removal does not affect the connectivity of $S \cup R$. Since at least $2k$ self loops are incident to r and none of them are used in any S -subgraph by construction, then r is balanced as well. Hence, we only need to balance $S \cup R'$ to achieve balancing on $S \cup R$. $\square$

Lemma 3.0.5. *$S \cup R$ is $(Q - 4)k$ -edge-connected in G .*

Proof. Assume not. Take the smallest edge cut of $S \cup R$, and note that it must break $S \cup R$ into $S \cup R_1$ and R_2 , since if it cut S , then it would contradict the statement that S is Qk -edge-connected.

Let the two components after the cut be C_1 which contains $S \cup R_1$ and C_2 which contains R_2 . Contract C_2 into a vertex v_2 . Note that C_2 cannot be a single vertex, as if it were a single vertex, this would imply that that vertex is in R and has at least $2k$ self loops, contradicting our assumption that every vertex in R has less than $2k$ self loops. Hence, $C_1 + v_2$ is strictly smaller than G . As $C_1 + v_2$ satisfies the properties of the extension theorem, we can find k edge disjoint S -subgraphs that balance $S \cup R_1$ and extend v . Now undo the contraction on C_2 . We wish to find a collection of edge disjoint paths $P_1, P_2, \dots, P_l$ where l is the size of the cut, such that each path starts in C_1 , traverses an edge of the cut as its very first edge, and all of the paths meet at a common vertex x in C_2 . This will ensure connectivity of the S -subgraphs in G , as we can just replace the edges incident to v_2 in the S -subgraphs in $C_1 + v_2$ with the respective paths when we undo the contraction.

To find such a vertex x let us first contract C_1 into v_1 and look at $C_2 + v_1$. Note that for any vertex $r \in R_2$, there must exist l edge disjoint paths between r and v_1 , because if there are not then this must mean there is an $S \cup R$ cut smaller than l , contradicting the fact that we took the minimum cut. Hence, we just need to pick r such that we ensure that the S -subgraphs will balance R_2 . We will do this by picking the r such that the sum of the lengths of the edge disjoint paths $P_1, P_2, \dots, P_l$ from v_1 to r_1 is minimized.

We claim that the degree of any $r \in R_2$ in $P_1 \cup P_2 \dots P_l$ does not exceed l . Suppose not, and there exists such r' that violates this. Note that firstly the degree of r' in $P_1 \cup P_2 \dots P_l$ is even as paths that pass through r' must contribute even degree. Note that each path also passes through r' once, because if not we can short-cut the path by skipping the portion where the path leaves r' for the first time and enters r' for the second time. Suppose l' paths pass through r' , and without loss of generality let them be $P_1, P_2, \dots, P_{l'}$. We can create a new collection of edge disjoint paths, $P'_1, P'_2, \dots, P'_l$ from v_1 to r' , where $P'_1, \dots, P'_{l'}$ are simply $P_1, P_2 \dots P_{l'}$, except we truncate them the first time they reach r' . Form $P'_{l'+1}, \dots, P'_l$ by taking $P_{l'+1}, \dots, P_l$. Note that when we truncated $P_1, P_2, \dots, P_{l'}$ into $P'_1, P'_2, \dots, P'_{l'}$, we are left with a collection of l' edge disjoint paths from r' to r unused. We can then use $l - l'$ of these to finish

$P'_{l'+1}, \dots, P'_l$ and connect them to r' . Note that since $2l' > l$, then $l' > l - l'$, which means not all of the remaining truncated paths will be used. Hence, we have formed $P'_1, P'_2, \dots, P'_l$, a collection of edge disjoint paths from v_1 to r' which uses a strict subset of edges in $P_1 \cup P_2 \dots P_l$, contradicting the fact that the sum of the lengths of $P_1 \cup P_2 \dots P_l$ is minimized.

Hence, no vertex $r \in R_2$ has degree more than $(Q - 4)k$ on $P_1 \cup P_2 \dots P_l$, which ensures that each vertex $r \in R_2$ has at least $2k$ unused edges and hence will be balanced. Setting $x = r$ and undoing the contraction gives us k edge disjoint subgraphs which connect S and balanced $S \cup R$, hence resolving the case where there is an edge cut of at most $(Q - 4)k$. $\square$

Lemma 3.0.6. $|S| \geq 3$

Proof. We note that this proof is basically the same as "When $|S| \leq 2$ " in Lau's thesis [3]. We consider the case where $|S| = 2$. In this case, S consists of v and another vertex u . Take any collection of Qk edge disjoint paths between v and u . Denote $P_i = \{P_{i,1}, P_{i,2}, \dots, P_{i,m_i}\}$ as the paths from v to u that start by traversing an edge with label i from v . Note that P_i has a size of at least 2 since we assumed each label appears twice at v .

Construct H_i by taking $P_{i,1} \cup P_{i,2}$. Then, for any remaining paths in P_i , iteratively add them to H_i by starting at v , traversing the path to u , and truncating immediately when they reach a vertex of H_i next, essentially adding an ear to H_i . Note that by this construction, u and v are connected, H_i remains connected even after v is deleted, and the degree of u and v in H_i is at least 2.

We now prove that R is balanced by claiming for each vertex $r \in R$, at least $2k$ edges go unused in $H_1, H_2, \dots, H_k$. Take a vertex $r \in R$, and assume it appears in l paths/ears in the construction of $H_1, H_2, \dots, H_k$. Note that by our construction, r has a degree of 2 in at most $2k$ of these paths/ears. This is because once we add the first two paths to H_i , we truncate future paths the first time they reach H_i . Hence, if r was already H_i , and we added another path to H_i from $\{P_{i,3}, P_{i,2}, \dots, P_{i,m_i}\}$, then if r was in that path it would either stop at r or before r , contributing a degree of at most 1.

If $l \leq 2k$, at most $4k$ edges were used at r , hence at least $2k$ edges remain unused.

Otherwise, when $l > 2k$, we can upper bound the degree of r in $H_1, H_2, \dots, H_k$ with $(l - 2k) + 4k = l + 2k$. Note that $l \leq d(r)/2$, where $d(r)$ denotes the number of edges incident to r , since $H_1 \cup H_2 \cup \dots \cup H_k$ is a subset of Qk edge disjoint paths between v and u . This gives us that at least $d(r) - (d(r)/2 + 2k) \geq ((Q - 2)/2 - 2)k$ edges remain unused at r , which is at least $2k$ as long as $Q \geq 10$. $\square$

Theorem 3.0.3 (DeVos, McDonald, Pivotto [5]). *Let G be a loopless multigraph, and let some $S \subseteq G$ where $|S| \geq 2$ be $2\lceil(5k + 3)/2\rceil$ -edge-connected. Suppose there exists some $v \in S$ with degree exactly $2\lceil(5k + 3)/2\rceil$. Then for any edge subpartition at v where at least k edges are unlabelled and each label appears once, there exist k edge-disjoint S -subgraphs that extend v and leave at least k edges unused at each vertex in S .*

Corollary 3.0.1. *Let G be an arbitrary loopless multigraph, and let S be a $5k + 4$ connected subset of $V(G)$. Then there exist k edge disjoint S -subgraphs that balance S .*

Proof. We can assume $|S| \geq 2$ otherwise this is trivial. Delete edges from G until S is minimally $2\lceil(5k + 3)/2\rceil$ -edge connected. Take a minimum edge cut Z' of S and break the remaining graph into components C_1 and C_2 . Condense C_1 and C_2 to v_1 and v_2 respectively, and examine $C_1 + v_2$ and $C_2 + v_1$. v_2 is clearly $2\lceil(5k + 3)/2\rceil$ -edge-connected to $S \cap C_1$, as there must be $2\lceil(5k + 3)/2\rceil$ -edge-disjoint paths from C_1 to C_2 . Note that $|Z'| = 2\lceil(5k + 3)/2\rceil$ since we assumed minimal connectivity. Using DeVos's extension theorem, we can pack k edge disjoint $(S \cap C_1) \cup v_2$ -subgraphs on $C_1 + v_2$ such that each vertex has at least k incident edges unused. We can then label the edges at v_1 accordingly, and apply the same extension theorem to $C_2 + v_1$ to get k edge disjoint $(S \cap C_2) \cup v_1$ -subgraphs that extend v_1 . Combine the packings by taking the edge-union of the i th subgraph on $C_2 + v_1$ and the i th subgraph on $C_1 + v_2$. We then obtain k edge disjoint S -subgraphs with k incident edges unused for each vertex in S , hence achieving balancing. $\square$

Let W be the neighbors of v that are not in $S \cup R$, and define $G' = G - v - W$. Note that G' retains the property that every $u \in V(G') - S - R$ has degree 3 and is adjacent to

exactly 3 vertices.

Lemma 3.0.7. *If $k \geq 8$, $(S \cup R - v)$ is at most $(\lfloor 5.5k \rfloor - 1)$ -connected in G' .*

Proof. Assume that $S \cup R - v$ in G' is $\lfloor 5.5k \rfloor$ -edge connected. Since $k \geq 8$, we know that $S \cup R - v$ in G' is at least $5k + 4$ edge-connected. Then, by the previous lemma, we can pack k edge disjoint $(S \cup R) - v$ subgraphs that balance $(S \cup R) - v$. Then extending v on G is easy, giving us k edge disjoint $S \cup R$ subgraphs which balance $S \cup R$ and extend v . $\square$

Let Z be the minimum $(S \cup R) - v$ edge cutset of G' , and note that $|Z| \leq 5.5k - 1$ when $k \geq 8$. We note that when $k < 8$, we can still claim that $|Z| < 6k$ as a consequence of the next two lemmas below. Following the rest of the proof with the assumption that $|Z| < 6k$ as opposed to $|Z| < 5.5k$ yields a bound of $36k$ as opposed to $35k$. The following lemma is proved in Lau's thesis, which is based on a result of Frank, Kiraly, and Kriesell [7].

Lemma 3.0.8 (Lau 3.7.9 [3]). *Let G be a graph and S be a subset of $V(G)$. Suppose S is $3k$ -edge-connected in G , every vertex in $V(G) - S$ is of degree 3, and there is no edge between vertices in $V(G) - S$. Let T be a set of edges with $|T| \leq k$. Then $G - T$ has k edge-disjoint S -subgraphs.*

Lemma 3.0.9. *$(S \cup R - v)$ is at most $(6k - 1)$ -connected in G' .*

Proof. Assume that $(S \cup R - v)$ in G' is $6k$ -edge connected. Note that every vertex in $G' - S - R$ is adjacent to exactly 3 different vertices in $S \cup R$ and has degree exactly 3. Indeed, note that all the edges between $v \cup W$ and G' are either from v to a vertex in $S \cup R$ or from a vertex in W which must only have neighbors in $S \cup R$, hence all vertices in $G' - S - R$ have the same edges and neighbors in G' and G . Since $G' - S - R$ is an independent set, this means that we can find $2k$ edge disjoint $(S \cup R - v)$ -connectors on G' . We can union two $(S \cup R - v)$ -connectors together to form an $(S \cup R - v)$ -subgraph where every vertex in $(S \cup R - v)$ has degree at least 2. Denote these graphs as $H_1, H_2, \dots, H_k$.

For the edges incident to v that have a label i , add that edge to H_i . If the other endpoint x of that edge is in W , then add both of the other two edges incident to x to H_i as well. Note

that since we assumed that each label appears at least twice at v , each of H_i now connects $S \cup R$ while extending its respective edges. Furthermore, the degree of all vertices in $S \cup R$ (including v) is at least 2 in H_i , ensuring that $H_1, H_2, \dots, H_k$ balance $S \cup R$, as desired. $\square$

Lemma 3.0.10. *$G' - Z$ has 2 connected components.*

Proof. Note that it is sufficient to prove that G' has at most two components. This is because if G' has exactly two components, $|Z|$ will be empty and this will trivially be true. Otherwise, if G' is connected, note that deleting a single edge can increase the number of connected components by at most 1, hence $G' - Z$ will have exactly two components by minimality of Z . Take any component of G' , and note that for each component, there must be a vertex that was incident to v or W on G . Since vertices in W are only incident to vertices in $S \cup R$ by Lemma 3.0.3, and all neighbors of v in G' are in $S \cup R$, each connected component must contain at least one vertex in $S \cup R$.

Suppose that there are at least three components in G' , and select u_1, u_2, u_3 each from a different component such that all three are vertices in $S \cup R$. Then there is a collection of $(Q - 4)k$ edge disjoint paths from u_i to v , take those paths and truncate them the first time they traverse to a neighbor of v . Since v has at most Qk neighbors in G , there must be some common vertex $w \in N_G(v)$ which u_1, u_2, u_3 all have a path to, as long as $Q \geq 13$. If $w \in S \cup R$, then u_1, u_2, u_3 are still all connected, which is a contradiction. Otherwise, those three paths must've traversed through a neighbor of w that is not v , but since $w \notin S \cup R$, then w must have three neighbors, including v . Hence, two paths will need to traverse through the same neighbor of w to reach w , and hence still remain connected in G' , a contradiction. $\square$

We denote the components as C_1 and C_2 . For each $w_i \in W$, note that it has exactly two neighbors x_i and y_i in G' , which we call a couple. Let X_1 be the collection of couples where both vertices are in C_1 , X_2 be the collection of couples where both vertices in X_2 , and let X_C be the collection of couples with one vertex in X_1 and another vertex in X_2 . Let B_i be $C_i \cap N(v) \cap (S \cup R)$, and let $c(B_i)$ denote the number of edges between v and B_i .

Definition 3.0.1. We say v_1 and v_2 have λ common paths if there exist λ edge disjoint paths starting from v_1 , λ edge disjoint paths starting from v_2 , and a bijection between the paths such that each pair of paths ends at the same vertex. We allow the empty path (from a vertex to itself) to count as a path, as long as the other vertex has a corresponding path to that vertex.

Lemma 3.0.11 (Lau 3.7.5 [3]). For any arbitrary graph G , if v_1 and v_2 have $2\lambda + 1$ common paths in G , there exist $\lambda + 1$ edge disjoint paths from v_1 to v_2 in G .

We first tackle the case where C_1 and C_2 both contain vertices in S .

Lemma 3.0.12. If both C_1 and C_2 contain vertices in S , $|X_C| \geq Qk - 2|Z|$.

Proof. Take some $u_1 \in S \cap C_1$ and note that there are Qk edge disjoint paths from v to u_1 in G . Note that there are $|X_2| + c(B_2)$ of these paths that start from v and go to C_2 before reaching u_1 . Hence, these paths must all pass through Z at some point, so $|Z| \geq |X_2| + c(B_2)$. Then, by a similar argument, $|Z| \geq |X_1| + c(B_1)$ as well. Note that $Qk = |X_C| + |X_1| + |X_2| + c(B_1) + c(B_2)$, giving us $Qk \leq |X_C| + 2|Z|$ and hence $|X_C| \geq Qk - 2|Z|$, as desired. $\square$

Lemma 3.0.13. Given a vertex in $x \in S \cup R - v$, there exists a collection $P(x)$ of $(Q - 4)k$ edge disjoint paths in G from v to x that use exactly one edge incident to v , and remain connected after deletion of v and W .

Proof. Note there must exist a collection $(Q - 4)k$ edge disjoint paths from v to x on G , and let $P(x)$ be such a collection of $(Q - 4)k$ edge disjoint paths such that the sum of lengths of the paths is minimized. Note that no path traverses through v twice by minimality. Now, assume, for contradiction, that such a path becomes disconnected after deletion of v and W . Then such a path must have traversed to a vertex in G' before then traversing to a vertex $w \in W$. Then vw could not have been used in $P(x)$, so we can shorten this path by cutting off the first part of the path that starts at v and traverses to G' and then to w , by simply replacing it with vw . Hence $P(x)$ exists for any $x \in S \cup R - v$, as desired. $\square$

For each vertex x , assign $P(x)$ to be a collection of paths as described in the previous lemma. Define $P'(x)$ to be the truncated collection of paths in $P(x)$ after the deletion of v and W , that is, $P'(x) = \{P - v - W | P \in P(x)\}$. Let S_1 and S_2 denote $S \cap C_1$ and $S \cap C_2$ respectively. Similarly, let R_1 and R_2 denote $R \cap C_1$ and $R \cap C_2$.

Lemma 3.0.14. *If C_1 and C_2 both contain vertices in S , then for any $a, b \in S_i \cup R_i$ where $a \neq b$, there exist $(Q - 19)k$ common paths between a and b in $G' - Z$. If we drop the assumption that Z splits S in G' , we can say that there exist $(Q - 23)k$ common paths between such a and b .*

Proof. We can assume, without loss of generality, that a and b are in C_1 . We aim to find common paths by using the neighbors of W in C_1 , and the endpoints of Z . Let k_a denote the crossing couples which $P'(a)$ does not have a corresponding path to, and let z_a denote the crossing couples which $P'(a)$ does have a corresponding path to, but the path traverses through Z (note that if a path in $P'(a)$ starts at a crossing couple but then goes to C_2 , it must pass through Z). Then a has $|X_C| - k_a - z_a$ paths to X_C in C_1 . Hence, a and b share at least $|X_C| - k_a - z_a - k_b - z_b$ common paths in C_1 which all end the crossing couples.

We now count the common paths a and b share through the endpoints in Z . Consider the paths in $P(a)$ that use an edge incident to v corresponding to X_2 or an edge incident to v that has its other endpoint in C_2 . All such paths must pass through Z , so truncate the paths the first time one would finish traversing Z if they started from a . There are $|X_2| + c(B_2) - m_a$ such paths, where $c(B_2)$ is the number of incident edges in v that has its other endpoint in C_2 , and m_a is for the number of such edges where a has no corresponding path to. We add back the z_a paths we ignored in the previous paragraph. Note that the expression $(|X_2| + c(B_2) - m_a + z_a) + (|X_2| + c(B_2) - m_b + z_b)$ counts the edges shared by $P'(a)$ and $P'(b)$ in Z exactly twice, and everything else less than twice, so $2|X_2| + 2c(B_2) - m_a - m_b + z_a + z_b - |Z|$ is a lower bound on the common paths between a and b that end in Z .

We have $(|X_C| - k_a - z_a - k_b - z_b) + (2|X_2| + 2c(B_2) - m_a - m_b + z_a + z_b - |Z|)$ as a lower bound on the common paths between a and b . Note that by construction of $P'(x)$,

$k_x + m_x \leq 4k$ for any $x \in S_1 \cup R_1$, so this simplifies down to $|X_C| + 2|X_2| + 2c(B_2) - 8k - |Z|$.

We now claim that $|X_C| + c(B_2) + |X_2| \geq Qk - |Z|$. Indeed, for any $s_1 \in S_1$ and $s_2 \in S_2$, there are Qk edge disjoint paths. s_2 can use at most $|Z|$ paths to reach s_1 without v , but otherwise needs to traverse through v or a crossing couple, having to go through an edge between X_C and C_2 , an edge between v and X_2 , or an edge between v and C_2 . Hence there are at least $Qk - |Z|$ such paths, implying the existence of at least $Qk - |Z|$ such edges. Note that if we drop the assumption that both C_1 and C_2 contain vertices from S , then our inequality becomes $|X_C| + c(B_2) + |X_2| \geq (Q - 4)k - |Z|$.

We now have that $|X_C| + 2|X_2| + 2c(B_2) - 8k - |Z| \geq Qk - 8k - 2|Z|$, and since $Z < 5.5k$, this gives us at least $(Q - 19)k$ common paths. In the latter case where Z may not split S , we obtain at least $(Q - 23)k$ common paths following the same computation. $\square$

Lemma 3.0.15. *If both C_1 and C_2 contain vertices in S , for each $i \in \{1, 2\}$ $S_i \cup R_i$ is $(\lfloor Q/2 - 9.5 \rfloor)k$ -edge-connected in $G' - Z$.*

Proof. Follows as a corollary from the previous lemma and Lemma 3.0.11. $\square$

Lemma 3.0.16. *If both C_1 and C_2 contain vertices in S , then G has k edge disjoint S subgraphs that balance $S \cup R$ and extend $P_k(v)$.*

Proof. We first handle the case where $|S_i \cup R_i| \geq 2$ for $i \in \{1, 2\}$. Note that if a vertex in $u \in V(G') - S - R$ is incident to an edge in Z , by minimality of Z , the other two edges must be contained in the same component C_i that u is in. To apply Lemma 3.0.8, we first replace ux_1 and ux_2 with x_1x_2 , which we note does not alter the connectivity nor the degree of $S_i \cup R_i$. We now have that every vertex in $C_i - S_i - R_i$ has degree 3 and exactly 3 neighbors.

Choose $\min(k, |Z|)$ edges in $|Z|$, and call them reserve edges. Note that each reserve edge may have one endpoint in $V(G') - S - R$. If such an endpoint u exists, then we delete the edge x_1x_2 we "split off" from the corresponding component C_i . Note that we delete at most k edges from each C_i . Call the altered component C'_i . Since all the deleted edges had both endpoints in $S_i \cup R_i$, and $(\lfloor Q/2 - 9.5 \rfloor) \geq 6$ as long as $Q \geq 31$, we can apply Lemma 3.0.8 and

find $2k$ edge disjoint $S_1 \cup R_1$ -subgraphs $A'_1, A'_2, \dots, A'_{2k}$ on C'_1 , and $2k$ edge disjoint $S_2 \cup R_2$ subgraphs $B'_1, B'_2, \dots, B'_{2k}$ on C'_2 . Set $A_1, A_2, \dots, A_k$ as $A_i = A'_i$. We claim that $A_1, A_2, \dots, A_k$ are $S_1 \cup R_1$ subgraphs that balance $S_1 \cup R_1$. Indeed, for each vertex $x \in S_1 \cup R_1$, there must be at least k unused edges at each vertex, because we can also add $A'_{k+1}, A'_{k+2}, \dots, A'_{2k}$ to the packing, and each of these subgraphs uses at least one edge incident to x . Similarly, we can construct $S_2 \cup R_2$ subgraphs $B_1, B_2, \dots, B_k$ of C_2 by setting $B_i = B'_i$, which will also balance $S_2 \cup R_2$. If $x_1x_2 \in A_j$ and was an edge we split off, delete x_1x_2 and add x_1u and x_2u to A_j , where u was the vertex we split off from. We repeat this for C_2 .

If $k \leq |Z|$, then we have k reserve edges. Label them $e_1, e_2, \dots, e_k$. If both endpoints of e_j are in $S \cup R - v$, then set $H_j = A_j \cup B_j + e_j$, which is connected. If e_j has an endpoint u not in $S \cup R$ and hence was previously split off, set $H_j = A_j \cup B_j + x_1u + x_2u$, where x_1u and x_2u were the edges split off to form e_j . Since each vertex in $V(G') - R - S$ has at most one edge in Z , there is no overlap between H_i and H_j where $i \neq j$, and H_j is connected, as desired. We then add back the vertices in W , and if vw had a label of j , we add the other two edges of w to H_j , giving us our desired result.

Now suppose $k > |Z|$. In this case, our reserve edges are not enough for connectivity, so we need to utilize the crossing couples. If vw where $w \in W$ has a label of j , and w is adjacent to x, y which form a crossing couple, then define $H_j = A_j \cup B_j + xw + yw$, which gives us the desired connectivity and extension properties. Now our only issue is the labels which do not correspond to crossing couples. Note that $|X_C| \geq Qk - 2|Z|$, and since each label appears twice at v , there can be at most $|Z|$ labels missing from the crossing couples. For each missing label, we then label the reserve edges, and repeat the previous procedure we did when $k \leq |Z|$ to obtain H_j . Hence this gives us $H_1, H_2, \dots, H_k$ which connect S , balance $S \cup R$, and extend v , as desired.

Now we handle the case where there exists a component that has exactly one vertex in $S \cup R$, without loss of generality let that component be C_1 . Note that C_1 consists of only one vertex in S , which we call s , because if that vertex was incident to some vertex $u \in V(G') - S - R$

which was also part of C_1 , then two edges incident to u would need to be in Z , but we could get a smaller cut by moving u to C_2 , contradicting the minimality of Z . Construct $B_1, B_2, \dots, B_k$ with the same method outlined previously, and we use exactly k reserve edges and crossing couples to connect s to $B_1, B_2, \dots, B_k$. For the remaining crossing couples, instead of adding both edges incident to C_1 and C_2 to the corresponding graph in $B_1, B_2, \dots, B_k$, only add the edge that is incident to C_2 . There are at least $Qk - 2(5.5k) = (Q - 11)k$ crossing couples, k of which we might have used to connect s to the rest of the graph, leaving at least k crossing couples unused at s , hence balancing s , as desired. $\square$

We now handle the case where only one component contains vertices in S . Without loss of generality, let that component be C_1 .

Lemma 3.0.17. *If $S \subseteq C_1$, for each $i \in \{1, 2\}$ $S_i \cup R_i$ is $(\lfloor Q/2 - 11.5 \rfloor)k$ -edge-connected.*

Proof. Follows as a corollary from Lemma 3.0.14. As there are $(Q - 23)k$ common paths between any $a, b \in S_i \cup R_i$, $S_i \cup R_i$ is $(\lfloor Q/2 - 11.5 \rfloor)k$ -edge-connected. $\square$

Lemma 3.0.18. *If C_2 only contains vertices in R , $|X_C| \geq (Q - 4)k - 2|Z|$.*

Proof. Take some $u_1 \in S \cap C_1$ and note that there are Qk edge disjoint paths from v to u_1 on G . Note that there are $|X_2| + c(B_2)$ of these paths that start from v and go to C_2 before reaching u_1 . Hence, these paths must all pass through Z at some point, so $|Z| \geq |X_2| + c(B_2)$.

Now, take some vertex $u_2 \in R \cap C_2$. Take a collection of $(Q - 4)k$ edge disjoint paths $P(u_2)$ on G from v to u_2 as defined in Lemma 3.0.13. Note that at least $|X_1| + c(B_1) - 4k$ paths of these $(Q - 4)k$ paths must start at C_1 , and all of them must have passed through Z to reach u_2 . This gives us $|Z| \geq |X_1| + c(B_1) - 4k$. Note that $Qk = |X_C| + |X_1| + |X_2| + c(B_1) + c(B_2)$, giving us $Qk \leq |X_C| + 2|Z| + 4k \rightarrow |X_C| \geq (Q - 4)k - 2|Z|$, as desired. $\square$

Lemma 3.0.19. *If $S \subseteq C_1$, then G has k edge disjoint S subgraphs that balance $S \cup R$ and extend $P_k(v)$.*

Proof. If $|Z| \geq k$, we repeat the proof in Lemma 3.0.16, where we label k edges in Z as reserve edges, and use them to join $A_1, A_2, \dots, A_k$ and $B_1, B_2, \dots, B_k$. Note that as long as $Q \geq 35$, as $\lfloor Q/2 - 11.5 \rfloor \geq 6$ allowing this proof to hold. Also note that there are at least $35k - 4k - 11k = 20k$ crossing couples, which ensures balancing in the event that one component consists of a single vertex from $S \cup R$.

We handle the case where $|Z| < k$. Let B be the neighbors of v in C_2 , and let W' be a subset of W which have two neighbors in C_2 . Note that B and W' are essentially what we need to extend through C_2 to reach C_1 . Fix any arbitrary vertex in $s \in S_1$. Note that on G , there are exactly Qk edge disjoint paths from v to s . Let $P_1, P_2, \dots, P_l$ be the paths that go through B or W' . Since these paths are edge disjoint and all of them must pass through Z in order to reach C_1 , we can assume that $l \leq |Z|$.

End the paths $P_1, P_2, \dots, P_l$ once they go through an edge in Z , and label all such edges as reserve edges. Split off all vertices of degree 2 in C_1 in $G' - Z$, and delete the edges corresponding to the reserve edges. We can again find edge disjoint $A_1, A_2, \dots, A_k$ which connect $S_1 \cup R_1$ and leave k edges unused at each vertex in $S_1 \cup R_1$ in C_1 . We then use the deleted edges to connect the reserve edges to $S_1 \cup R_1$, hence properly extending all edges of v with one endpoint in B or W' .

We add back the vertices in $W - W'$, and if a vertex forms a crossing couple, we use the edge that connects it to C_1 for extension. Then add back the remaining edges of v , which should all have their endpoints in C_1 , to get our final packing of $H_1, H_2, \dots, H_k$. Note that we only need to check that R_2 is balanced, and by construction, the only edges we used incident to R_2 were to construct $P_1, P_2, \dots, P_l$. Since there are at most k paths, each vertex in R_2 used at most $2k$ incident edges, and hence there are clearly at least $2k$ edges not used for each vertex in R_2 , as desired. $\square$

This concludes the proof of the Extension Theorem.

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